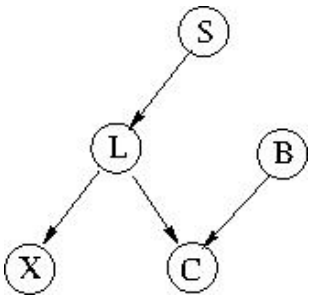
FIT3080 Artificial Intelligence
Practice Questions for Final Exam – Bayesian Networks

1. **[Bayesian Reasoning]** Consider the following Bayesian network for another version of the medical diagnosis example, where *B=Bronchitis, S=Smoker, C=Cough, X=Positive X-ray* and *L=Lung cancer*, and all nodes are Boolean. Suppose that the prior for a patient being a smoker is 0.25, and the prior for the patient having bronchitis is 0.05. List the pairs of nodes that are conditionally independent in the following situations, and explain why.



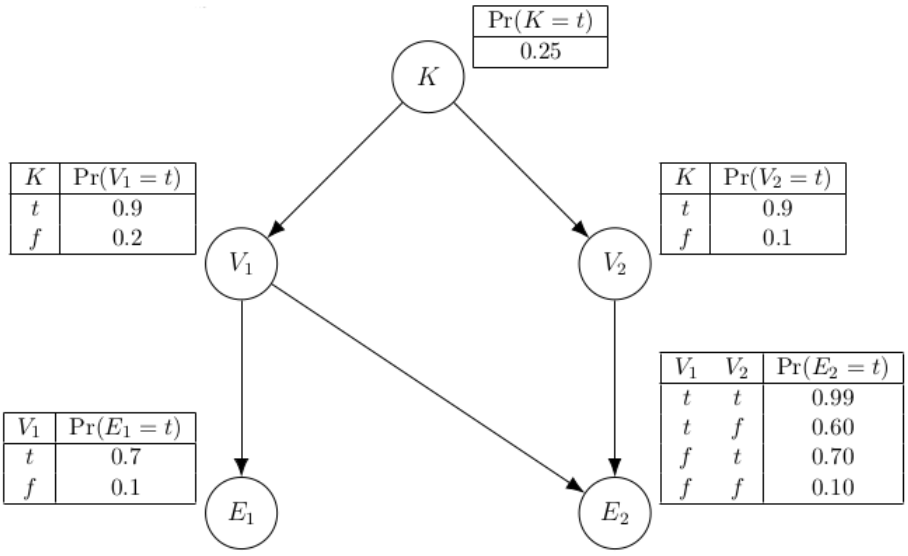
   (a) There is no evidence for any of the nodes.

   (b) The lung cancer node (L) is set to true (and there is no other evidence).

   (c) The smoker node (S) is set to true (and there is no other evidence).

   (d) The cough node (C) is set to true (and there is no other evidence).

2. **[Bayesian Networks]** Consider the following Bayesian Network. All variables are Boolean, so they can take either True (t) or False (f) as their values.



   Calculate the value of $Pr(K = t, V_1 = f, V_2 = t, E_1 = f, E_2 = t)$. Show the formula and the instantiated values in the formula.

3. [**Bayesian Networks**] Consider the following scenario about computer systems:

> *A* **system crash** *(***SC***) may be caused by an* **operating system failure** *(***OSF***), a* **hardware failure** *(***HF***) and/or* **malware** *(***M***). An* **OSF** *in turn may be caused by* **M** *and/or by a* **power failure** *(***PF***), which may be caused by* **stormy weather** *(***SW***). Finally, a* **PF** *may cause a* **HF**.

(a) Design a BN that represents this situation, including the conditional probability formulas for each node, e.g., $\Pr(X|Y,Z)$.

Notes:

- The boldfaced concepts should be represented in the BN.
- All the variables are binary (they have two values).
- You don't need to specify the actual probabilities in the CPTs.

(b) Based on the given scenario and the semantics of the nodes, could M, OSF and HF be in an Explain Away relation? Justify your answer using the features required for an Explain Away relation to hold.

4. [**Bayesian Decision Networks**] Consider a student who has the choice to buy or not buy a textbook for a course. We'll model this as a decision problem with one Boolean decision Node, B, indicating whether the student chooses to buy the book, and two Boolean chance nodes, M, indicating whether the student has mastered the material in the book, and P, indicating whether the student passes the course. Of course, there is also an *additive* utility node, U. A certain student, Sam, has a utility function: 0 for not buying the book and -$100 for buying it; and $2000 for passing the course and 0 for not passing. In this case, the additive utility for passing and buying the book is $2000 - 100$.

Sam's conditional probability estimates are as follows:
$Pr(p|b,m) = 0.9 \quad Pr(m|b) = 0.9$
$Pr(p|b,\neg m) = 0.5 \quad Pr(m|\neg b) = 0.7$
$Pr(p|\neg b,m) = 0.8$
$Pr(p|\neg b,\neg m) = 0.3$
You might think that P would be independent of B given M, but this course has an open-book final – so having the book helps.

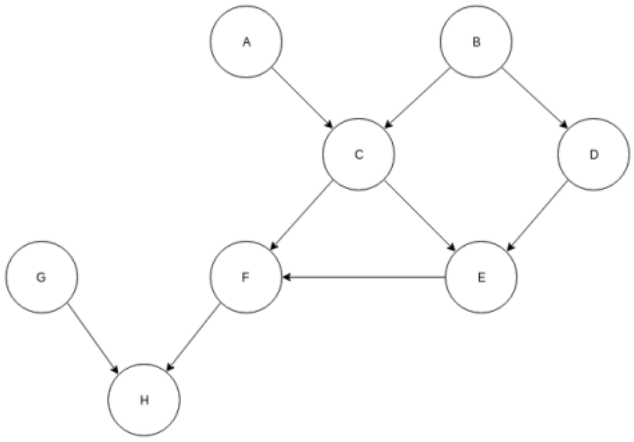
(a) Draw the decision network for this problem.

(b) Compute the expected utility of buying the book and of not buying it.

(c) What should Sam do?

5. [**D-separation**] According to this Figure, determine whether the following claims to

be True or False and justify your answer.



(a) $B \perp G|A$

(b) $C \perp D|F$

(c) $C \perp D|A$

(d) $H \perp B|C, F$